Water Analysis

MACHEREY-NAGEL

Compact photometer PF-3



The smart solution for special applications

- Intuitive menu guidance
- USB connection and free software
- Add new tests anytime for free
- Lightweight and robust



Compact photometer PF-3

Addition to the MN family

The compact photometer PF-3 is the smallest member of the MACHEREY-NAGEL photometer family and perfectly fits our tradition of reliability, user friendliness and innovation.

The following diagram highlights the available versions and respective parameters.



The PF-3 is equipped with three LEDs emitting three different wavelengths, which are independent for each version. These versions are designed to meet the analysis requirements of specific applications. Currently, we can provide five distinct PF-3 versions for pool, drinking water, soil, COD analysis and fishing waters. Each version can analyze a particular set of parameters, which are vital to analyze in the respective area of application. All versions come in three different options including a box with the photometer, a rugged case with photometer and accessories, or a case with pre-packed reagents.



Compact photometer PF-3

Test platforms for the PF-3

The PF-3 can handle both our convenient *NANOCOLOR* ® tube tests, as well as the economical *VISOCOLOR* ® test kits.

Combining both platforms within, the instrument provides maximum flexibility and allows customers to address their individual analysis needs.



NANOCOLOR® tube tests

Our barcoded tube tests are made of premium glass ware with an extra wide 16 mm tube opening. The accurately pre-dosed reagents cover the range of a wide variety of water and waste water parameters. The reaction methods are based on international standards (EPA, APHA, ISO, …) and provide a large portfolio with multiple measuring ranges, that can be run on all our photometers.

- Convenient handling
- Highest accuracy and safety
- Precise and reliable results

VISOCOLOR® *ECO* test kits

Our *VISOCOLOR*® *ECO* test kits come as colorimetric or titrimetric tests and provide the basic range of water analysis tests in our portfolio. These high quality tests are packed with all necessary reagents as well as premium color charts in only one box for easy ordering. Fast dissolving reagents and the easy dosage save time and allow easy working procedures. Most colorimetric kits can be read photometrically on our compact photometers PF-12*Plus* and PF-3.



- Outstanding cost-value ratio
- Time saving
- Effortless operations

VISOCOLOR® Powder Pillows

VISOCOLOR® Powder Pillows are photometric tests that combine easiest dosing of reagents with photometric precision. Each *VISOCOLOR*® Powder Pillow contains the exactly needed amount of reagents for a determination. *VISOCOLOR*® Powder Pillows can be evaluated on the compact photometers PF-12*Plus* and PF-3.



- Photometric precision
- For MN and competitor´s instruments
- Long shelf life



Compact photometer PF-3

Customer benefits at a glance

The PF-3 is designed with the customer in mind, providing a reliable and rugged solution for water analysis right at the point of interest. While being a tough compact photometer, the PF-3 is nevertheless packed to the brim with clever technical details, making it the most advanced instrument in its class.

At the core of the PF-3 are premium high precision optics with especially selected LEDs and first-rate interference filters. Additionally, the PF-3 impresses with intuitive handling, a fully developed menu structure, as well as various power supply options and free of charge software



Small and compact

- Handy and easy to carry
- Light and mobile
- Compact dimensions

Ideal for all your analysis needs in the lab and in the field



Easy and convenient

- Intuitive operation using just 4 buttons
- Fully developed menu structure
- Bright display
- Open cuvette slot

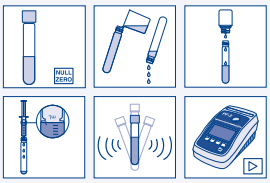
Convenient operation without the need for time consuming training



Strong and tough

- Glass-fiber reinforced housing
- Water proof according to IP 68
- Shock-resistant optics

Safe results even under the most difficult conditions



Comfortable and fast

- Intuitive instructions with pictograms
- Color-coded reagent bottles with clear dosing instructions
- Fast-dissolving reagents save time
- Maximum safety for the user and easy disposal

Convenient handling for quick analysis

Good to know

Versatile reagent case solutions for the PF-3 allow individual combination with test kits.

Compact photometer PF-3



Safe and flexible

- Runs on standard or rechargeable batteries
- Power supply via USB port
- Optional internal accu-pack chargeable via USB

Consistent and safe workflow



Individual and adaptive

- Various case solutions including reagents
- Rugged cases with premium foam inlays
- Flexible case content for multiple applications

Convenient operation with the perfect case for your need



Smart and clever

- Evaluation of *VISOCOLOR® ECO* tests, *VISOCOLOR®* Powder Pillows and *NANOCOLOR®* tube tests
- Results in different units
- Optional storage for just one zero-measurement
- Data transfer via USB

High measurement safety and easy data administration



Fair and economical

- Free of charge data export software
- Add new tests and parameters anytime
- Cost efficient photometer and reagent bundles
- Create customized methods for individual user

Consistently fair concept with impressive price-performance ratio

Compact photometer PF-3

For swimming pools



Swimming pools are important areas of recreation and relaxation. To constantly provide highest water quality and a safe user environment, the water within swimming pools has to be treated, conditioned and closely monitored. For reliable and easy water analysis for swimming pools, the PF-3 is the perfect tool.

The built-in LEDs allow quality control personnel to analyze the classic swimming pool parameters chlorine, pH, cyanuric acid and total alkalinity, as well as a selection of additional parameters, thus providing fundamental and vital information on pool water quality.

Tests suitable for the PF-3 Pool

Tests	Measurement range	Test number	No. of tests	REF
VISOCOLOR® ECO tests				
■ Alkalinity TA	5–250 mg/L $CaCO_3$	5-04	100	931204
■ Bromine	0.10–13.00 mg/L Br_2	5-11	200	931211
■ Chlorine 1, free, total and combined	0.05–2.00 mg/L Cl_2	5-35	150	931235
■ Chlorine 2, free, total and combined*	0.05–2.00 mg/L Cl_2	5-15	150	931215
■ free Chlorine 2*	0.05–2.00 mg/L Cl_2	5-16	150	931216
■ Chlorine 6, free, total and combined	0.05–6.00 mg/L Cl_2	5-17	200	931217
■ free Chlorine 6	0.05–6.00 mg/L Cl_2	5-19	400	931219
■ Chlorine dioxide*	0.10–3.80 mg/L ClO_2	5-21	150	931221
■ Cyanuric acid *	10–100 mg/L Cya	5-23	100	931223
■ Fluoride*	0.1–2.0 mg/L F^-	5-27	150	931227
■ Iron 1*	0.04–2.00 mg/L Fe	5-25	150	931225
■ Iron 2	0.04–2.00 mg/L Fe	5-26	150	931226
■ pH 6.0–8.2	6.10–8.40	5-70	150	931270
■ Silica HR 200*	10–200 mg/L SiO_2	5-34	100	931234
VISOCOLOR® Powder Pillows				
■ Chlorine, free	0.03–6.00 mg/L Cl_2	7-20	100 1000	936220 936220.1
■ Chlorine, total	0.03–6.00 mg/L Cl_2	7-21	100 1000	936221 936221.1
■ pH	pH 6.2–8.2	7-22	100	936222
■ Silica LR*	0.02–2.10 mg/L SiO_2	7-24	100	936224
■ Silica HR*	2–210 mg/L SiO_2	7-25	100	936225
■ Sulfate*	15–200 mg/L SO_4^{2-}	7-23	100	936223
NANOCOLOR® tests				
■ Chlorine / Ozone	0.05–2.50 mg/L Cl_2 / 0.05–2.00 mg/L O_3	0-17	20	985017
■ pH 6.5–8.2	6.10–8.40	0-72	100	91872



Ordering information

Description	REF
Compact photometer PF-3 Pool, in a cardboard box, incl. manual, batteries and certificate	919340
Compact photometer PF-3 Pool, in a case with foam inlay, incl. manual, batteries and certificate	934102
Reagent case with photometer PF-3 Pool, with *VISOCOLOR® ECO* reagents* chlorine 2 (liquid) free and total, pH 6.0–8.2, cyanuric acid, alkalinity TA	934118
Reagent case with photometer PF-3 Pool, with *VISOCOLOR® ECO* reagents* chlorine 6 (solid) free and total, pH 6.0–8.2, cyanuric acid, alkalinity TA	934119

* This product contains harmful substances which must be specially labeled as hazardous. For detailed information please see SDS.

Compact photometer PF-3

For drinking water

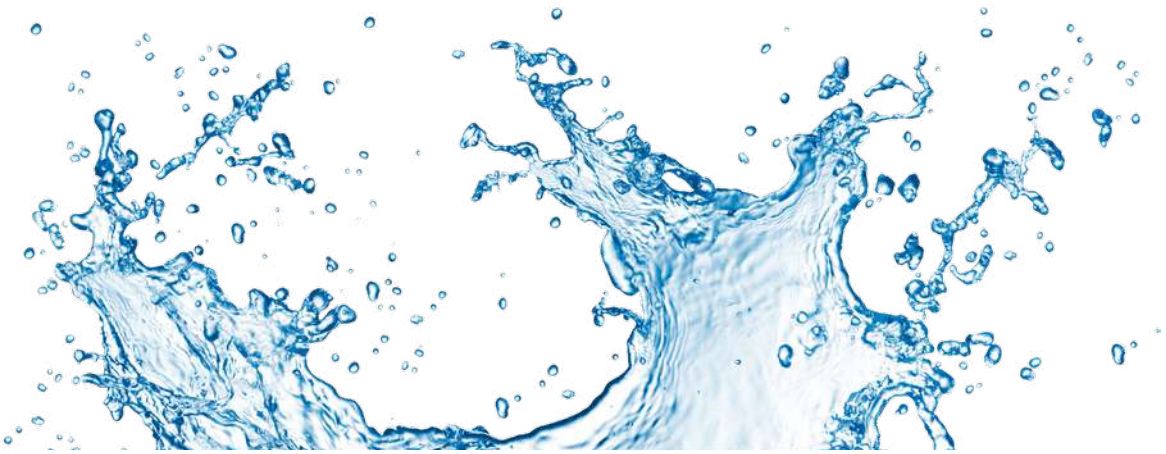


Drinking water is the most important foodstuff and cannot be replaced. Hence, it has to be wholesome and free of impurities that may adversely affect human health. Continuous water analysis is therefore the fundament for a sustainable and reliable drinking water supply.

The compact photometer PF-3 in combination with the proven *VISOCOLOR*® and *NANOCOLOR*® tests allows a quick and easy way of monitoring and checking the most important drinking water parameters (chlorine, chlorine dioxide, pH, fluoride and iron) for all water suppliers and industrial users during drinking water treatment.

Tests suitable for the PF-3 Drinking Water

Tests	Measurement range	Test number	No. of tests	REF
VISOCOLOR® *ECO* tests				
■ Alkalinity TA	5–250 mg/L $CaCO_3$	5-04	100	931204
■ Bromine	0.10–13.00 mg/L Br_2	5-11	200	931211
■ Chlorine 1, free, total and combined	0.05–2.00 mg/L Cl_2	5-35	150	931235
■ Chlorine 2, free, total and combined*	0.05–2.00 mg/L Cl_2	5-15	150	931215
■ free Chlorine 2*	0.05–2.00 mg/L Cl_2	5-16	150	931216
■ Chlorine 6, free, total and combined	0.05–6.00 mg/L Cl_2	5-17	200	931217
■ free Chlorine 6	0.05–6.00 mg/L Cl_2	5-19	400	931219
■ Chlorine dioxide*	0.10–3.80 mg/L ClO_2	5-21	150	931221
■ Cyanuric acid *	10–100 mg/L Cya	5-23	100	931223
■ Fluoride*	0.1–2.0 mg/L F^-	5-27	150	931227
■ Iron 1*	0.04–2.00 mg/L Fe	5-25	150	931225
■ Iron 2	0.04–2.00 mg/L Fe	5-26	150	931226
■ pH 6.0–8.2	6.10–8.40	5-70	150	931270
■ Silica HR 200*	10–200 mg/L SiO_2	5-34	100	931234
VISOCOLOR® Powder Pillows				
Chlorine, free	0.03–6.00 mg/L Cl_2	7-20	100 1000	936220 936220.1
Chlorine, total	0.03–6.00 mg/L Cl_2	7-21	100 1000	936221 936221.1
pH	pH 6.2–8.2	7-22	100	936222
Silica LR*	0.02–2.10 mg/L SiO_2	7-24	100	936224
Silica HR*	2–210 mg/L SiO_2	7-25	100	936225
Sulfate*	15–200 mg/L SO_4^{2-}	7-23	100	936223
NANOCOLOR® tests				
■ Chlorine / Ozone	0.05–2.50 mg/L Cl_2 / 0.05–2.00 mg/L O_3	0-17	20	985017
■ pH 6.5–8.2	6.10–8.40	0-72	100	91872



Ordering information

Description	REF
Compact photometer PF-3 Drinking Water, in a cardboard box, incl. manual, batteries and certificate	919343
Compact photometer PF-3 Drinking Water, in a case with foam inlay, incl. manual, batteries and certificate	934402
Reagent case with photometer PF-3 Drinking Water, with *VISOCOLOR*® *ECO* reagents* chlorine 2 (liquid), free and total, pH 6.0–8.2, fluoride, iron 2, chlorine dioxide	934124
Reagent case with photometer PF-3 Drinking Water, with *VISOCOLOR*® *ECO* reagents* chlorine 6 (solid), free and total, pH 6.0–8.2, fluoride, iron 2, chlorine dioxide	934125

* This product contains harmful substances which must be specially labeled as hazardous. For detailed information please see SDS.

Compact photometer PF-3

For soil analysis



Monitoring and reliable analysis is a corner stone in supporting and maintaining healthy, productive and biologically active soil. To effectively and efficiently plan all measures that affect the soil (fertilization, liming, etc.) it is crucial to determine the important soil parameters (nitrate, phosphate, potassium and ammonium). The PF-3 Soil in combination with the proven *VISOCOLOR® ECO* and *NANOCOLOR®* tests are ideally suited for mobile analysis of the soil quality right at the point of interest.

Tests suitable for the PF-3 Soil

Tests	Measurement range	Test number	No. of tests	REF
***VISOCOLOR® ECO* tests**				
■ Ammonium 3*	0.2–4.0 mg/kg NH_4-N	5-08	50	931208
■ Potassium*	40–300 mg/kg K^+	5-32	60	931232
■ Nitrate	2–28 mg/kg NO_3-N	5-41	110	931241
■ Phosphate*	20–500 mg/kg PO_4-P	5-84	80	931284
***NANOCOLOR®* tests**				
■ Ammonium 3*	0.08–4.60 mg/kg NH_4-N	0-03	20	985003
■ Ammonium 10*	0.4–16.0 mg/kg NH_4-N	0-04	20	985004
■ Ammonium 50*	2.0–80.0 mg/kg NH_4-N	0-05	20	985005
■ Potassium 50*	40–1000 mg/kg K^+	0-45	20	985045
■ Nitrate 50*	1–44 mg/kg NO_3-N	0-64	20	985064
■ Phosphate 5*	4–100 mg/kg PO_4-P	0-81	20	985081
■ Phosphate 15*	6–300 mg/kg PO_4-P	0-80	20	985080



Ordering information

Description	REF
Compact photometer PF-3 Soil, in a cardboard box, incl. manual, batteries and certificate	919341
Compact photometer PF-3 Pool, in a case with foam inlay, incl. manual, batteries and certificate	934202
Reagent case with photometer PF-3 Soil, with *VISOCOLOR® ECO* reagents* nitrate, potassium, ammonium 3, phosphate	934210
Reagent case for soil analysis with photometer PF-3 Soil, incl. reagents and accessories for sample preparation*	934220

* This product contains harmful substances which must be specially labeled as hazardous. For detailed information please see SDS.

Compact photometer PF-3

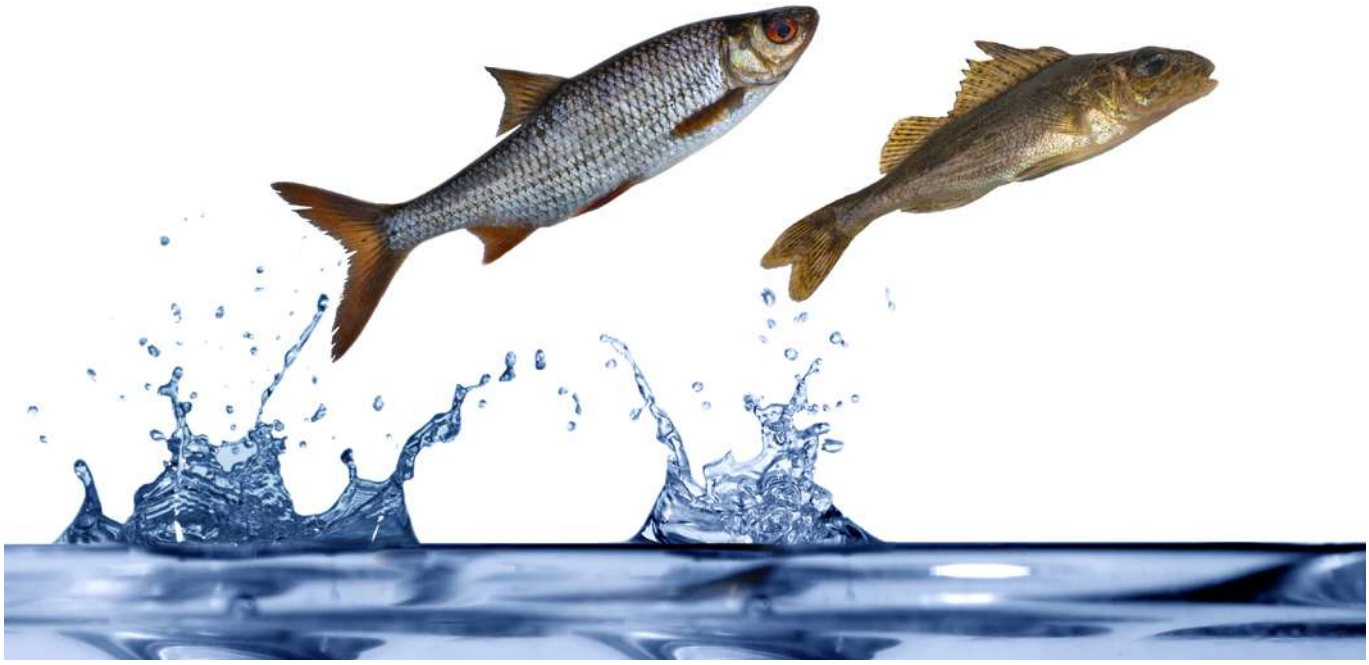
For fishing waters

A regular water monitoring is the precondition for a healthy and productive living space for fishes. To avoid deficits and fish mortality, it is important to determine all relevant parameters and react promptly.

With it´s three wavelengths, the compact photometer PF-3 Fish is perfectly suited for the analysis of fishing waters. Together with the proven *VISOCOLOR*® *ECO* and *NANOCOLOR*® tests, the PF-3 Fish is the perfect tool for a fast and reliable statement on the water quality.

Tests suitable for the PF-3 Fish

Tests	Measurement range	Test number	No. of tests	REF
VISOCOLOR® *ECO* tests				
■ Ammonium 3*	0.1–2.0 mg/L NH_4-N	5-08	50	931208
■ Chlorine 6 free, total and combined	0.05–6.00 mg/L Cl_2	5-17	200	931217
■ Iron 1*	0.04–2.00 mg/L Fe	5-25	150	931225
■ Iron 2	0.04–2.00 mg/L Fe	5-26	150	931226
■ Silica*	0.2–3.0 mg/L SiO_2	5-33	80	931233
■ Copper	0.1–5.0 mg/L Cu^{2+}	5-37	100	931237
■ Nitrate	1.0–14.0 mg/L NO_3-N	5-41	110	931241
■ Nitrite	0.01–0.15 mg/L NO_2-N	5-44	120	931244
■ pH 6.0-8.2	6.10–8.40 pH	5-70	150	931270
■ Phosphate*	0.2–5.0 mg/L PO_4-P	5-84	80	931284
■ Oxygen*	1–8 mg/L O_2	5-88	50	931288
NANOCOLOR® tests				
■ Ammonium 3*	0.04–2.30 mg/L NH_4-N	0-03	20	985003



Ordering information

Description	REF
Compact photometer PF-3 Fish, in a cardboard box, incl. manual, batteries and certificate	919345
Compact photometer PF-3 Fish, in a case with foam inlay, incl. manual, batteries and certificate	934602
Reagent case with compact photometer PF-3 Fish, incl. *VISOCOLOR*® *ECO* reagents, *VISOCOLOR*® *HE* reagents, QUANTOFIX® test strips and manual, batteries, certificate and accessories*	934127

* This product contains harmful substances which must be specially labeled as hazardous. For detailed information please see SDS.

Compact photometer PF-3

For COD analysis



COD levels play a major role in assessing communal and industrial waste water. As a sum parameter indicating the organic load of waste water, COD is one of the most important parameters in waste water analysis. Among others, COD values are used to calculate and monitor the purification capacities of waste water plants. The PF-3 COD together with the user friendly *NANOCOLOR*® tests is ideally suited to obtain information on the COD content (2–60000 mg/L) easily and directly at the place of sampling.

Tests suitable for the PF-3 COD

Tests		Measurement range	Test number	No. of tests	REF
NANOCOLOR® test					
■ COD 40*	ISO 15705	2–40 mg/L O_2	0–27	20	985027
■ COD 60*	ISO 15705	5–60 mg/L O_2	0–22	20	985022
■ COD 160*	ISO 15705	15–160 mg/L O_2	0–26	20	985026
■ COD 600*	ISO 15705	50–600 mg/L O_2	0–30	20	985030
■ COD 1500*	ISO 15705	100–1500 mg/L O_2	0–29	20	985029
■ COD LR 150*	ISO 15705	3–150 mg/L O_2	0–36	20	985036
■ COD HR 1500*	ISO 15705	20–1500 mg/L O_2	0-38	20	985038
■ COD 4000*		400–4000 mg/L O_2	0–11	20	985011
■ COD 10000*		1.00–10.00 g/L O_2	0–23	20	985023
■ COD 15000*		1.0–15.0 g/L O_2	0–28	20	985028
■ COD 60000*		5.0–60.0 g/L O_2	0–12	20	985012



Ordering information

Description	REF
Compact photometer PF-3 COD, in a box, incl. manual, batteries and certificate	919342
Compact photometer PF-3 COD, in a case with foam inlay, incl. manual, batteries and certificate	934302
NANOCOLOR® analysis case with compact photometer PF-3 COD, with additional space for 1 heating block *NANOCOLOR*® *VARIO C2* or *NANOCOLOR*® *VARIO Mini*, 2 pipettes, 3 *NANOCOLOR*® tube tests and accessories	919212

* This product contains harmful substances which must be specially labeled as hazardous. For detailed information please see SDS.

Compact photometer PF-3

VISOCOLOR® *ECO* and *NANOCOLOR*® tests suitable for the PF-3

VISOCOLOR® *ECO* tests		Measurement range	Test number	Version	No. of tests	REF
Alkalinity TA		5–250 mg/L $CaCO_3$	5-04	■■	100	931204
Ammonium 3*		0.1–2.5 mg/L NH_4^+ [1)]	5-08	■■	50	931208
Bromine		0.10–13.00 mg/L Br_2	5-11	■■	200	931211
Chlorine 1, free, total and combined		0.05–2.00 mg/L Cl_2	5-35	■■	150	931235
Chlorine 2, free, total and combined*		0.05–2.00 mg/L Cl_2	5-15	■■	150	931215
free Chlorine 2*		0.05–2.00 mg/L Cl_2	5-16	■■		931216
Chlorine 6, free, total and combined		0.05–6.00 mg/L Cl_2	5-17	■■■	200	913217
free Chlorine 6		0.05–6.00 mg/L Cl_2	5-19	■■	400	931219
Chlorine dioxide*		0.10–3.80 mg/L ClO_2	5-21	■■	150	931221
Copper		0.1–5.0 mg/L Cu^{2+}	5-37	■	100	931237
Cyanuric acid*		10–100 mg/L Cya	5-23	■■	100	931223
Fluoride*		0.1–2.0 mg/L F^-	5-27	■■	150	931227
Iron 1*		0.04–2.00 mg/L Fe	5-25	■■■	150	931225
Iron 2		0.04–2.00 mg/L Fe	5-26	■■■	150	931226
Nitrate		1.0–14.0 mg/L NO_3-N [1)]	5-41	■	100	931241
Nitrite		0.01–0.15 mg/L NO_2-N	5-44	■	120	931244
Oxygen*		1–8 mg/L O_2	5-88	■	50	931288
pH 6.0–8.2		6.10–8.40	5-70	■■	150	931270
Phosphate*		0.2–5.0 mg/L PO_4-P [1)]	5-84	■■	80	931284
Potassium*		2–15 mg/L K^+ [1)]	5-32	■	60	931232
Silica*		0.2–3.0 mg/L SiO_2	5-33	■	80	931233
Silica HR 200*		10–200 mg/L SiO_2	5-34	■■	100	931234
***VISOCOLOR*® Powder Pillows**						
Chlorine, free		0.03–6.00 mg/L Cl_2	7-20	■■	100 1000	936220 936220.1
Chlorine, total		0.03–6.00 mg/L Cl_2	7-21	■■	100 1000	936221 936221.1
pH		pH 6.2–8.2	7-22	■■	100	936222
Silica LR*		0.02–2.10 mg/L SiO_2	7-24	■■	100	936224
Silica HR*		2–210 mg/L SiO_2	7-25	■■	100	936225
Sulfate*		15–200 mg/L SO_4^{2-}	7-23	■■	100	936223
***NANOCOLOR*® tests**						
Ammonium 3*		0.05–3.00 mg/L NH_4^+ [1)]	0-03	■■	20	985003
Ammonium 10*		0.20–10.0 mg/L NH_4^+ [1)]	0-04	■	20	985004
Ammonium 50*		1.0–50.0 mg/L NH_4^+ [1)]	0-05	■	20	985005
Chlorine / Ozone		0.05–2.50 mg/L Cl_2 / 0.05–2.00 mg/L O_3	0-17	■■	20	985017
COD 40*	ISO 15705	2–40 mg/L O_2	0-27	■	20	985027
COD 60*	ISO 15705	5–60 mg/L O_2	0-22	■	20	985022
COD 160*	ISO 15705	15–160 mg/L O_2	0-26	■	20	985026
COD 600*	ISO 15705	50–600 mg/L O_2	0-30	■	20	985030
COD 1500*	ISO 15705	100–1500 mg/L O_2	0-29	■	20	985029
COD LR 150*	ISO 15705	3–150 mg/L O_2	0–36	■	20	985036
COD HR 1500*	ISO 15705	20–1500 mg/L 02	0-38	■	20	985038
COD 4000*		400–4000 mg/L O_2	0-11	■	20	985011
COD 10000*		1.00–10.00 g/L O_2	0-23	■	20	985023
COD 15000*		1.0–15.0 g/L O_2	0-28	■	20	985028
COD 60000*		5.0–60.0 g/L O_2	0-12	■	20	985012
Nitrate 50*		0.3–22.0 mg/L NO_3-N [1)]	0-64	■	20	985064
pH 6.5–8.2		6.10–8.40	0-72	■■	100	91872
ortho and total Phosphate 5*		0.20–5.00 mg/L PO_4-P [1)]	0-81	■	20	985081
ortho and total Phosphate 15*		0.30–15.00 mg/L PO_4-P [1)]	0-80	■	20	985080
Potassium 50*		2–50 mg/L K^+ [1)]	0-45	■	20	985045

* This product contains harmful substances which must be specially labeled as hazardous. For detailed information please see SDS.

[1)] For soil specific measurement ranges please refer to page 8 of the selection guide or *www.mn-net.com/PF*-3.

Additional tests and versions will be launched successively.
You can find all your current options at *www.mn-net.com/PF*-3.

Compact photometer PF-3

Technical data

PF-3	
Type	LED photometer with microprocessor control, self-test and auto calibration
Optics	LED + interference filters Insensitive to external light for quick measurements without cuvette slot cover
Wavelengths	3 wavelengths, depending on version Pool / Drinking Water: 450 nm / 530 nm / 590 nm Soil: 365 nm / 450 nm / 660 nm COD: 365 nm / 450 nm / 595 nm Fish: 450 nm / 530 nm / 660 nm
Wavelength accuracy	± 2 nm, bandwidth at half transmission 10 nm–12 nm
Light source	LED
Detector	Silicon photodiode
Measuring modes	*NANOCOLOR*® tube tests, *VISOCOLOR*® *ECO* tests and *VISOCOLOR*® Powder Pillows
Cuvette slot	Tubes 16 mm OD
Memory	50 results
Display	Backlit graphic display, 128 x 64 pix, all important data at a glance: result with unit, date, time
Operation	Self-explanatory menu guidance, plastic foil keypad, test selection via parameter lists
Interface	Mini-USB
Update	Free of charge via internet / PC
Operating range	10 °C–40 °C, up to 80 % relative humidity (non-condensing)
Power supply	3 AA batteries, rechargeable batteries, USB interface; optional internal accu-pack
Housing	Shock-resistant, waterproof and dustproof, according to IP68
Dimensions	170 mm x 95 mm x 68 mm
Weight	0.5 kg
Warranty	2 years
CE	CE-certified

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